Homoenergetic Landau equation: Well-posedness vs blow-up

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The Landau equation

We study a class of solutions to inhomogeneous **Landau** equation:

$$(\partial_t + v\cdot\nabla_x)f = \nabla_v \cdot \int_{\mathbb{R}^3} |v-v_*|^\gamma \Phi(v-v_*)\nabla_{v-v_*} f(x,v) f(x,v_*)\, dv_*,$$

where $\Phi(z) = |z|^2 I - z\otimes z$, and $\gamma \geq -3$. The value of γ greatly influences the behavior of solutions:

Value of γ	$[-3,-2]$	$(-2,1)$	$(\sqrt{3},2]$
Global existence space homogeneous	✓ [GS,'25]	✓ [GS,'25]	✓ [GS,'25]
Global existence near equilibrium	✓ [G,'02]	✓ [G,'02]	✓ [G,'02]
Global existence near vacuum	?	✓ [L,'19], [S,'23]	?
Finite-time singularities	?	?	✓ [BCGJVY,'26]

Theorem 1: Global well-posedness

Let $\gamma \in [-3,1]$ and $A \in \mathbb{R}^{3\times 3}$ with $\det(I+tA) > 0$ for $0 \leq t \leq T$. Let $g_{\text{in}}(\cdot) \geq 0$, $g_{\text{in}} \in C_c^\infty(\mathbb{R}^3)$ and initial data of the form

$$f_{\text{in}}(x,v) = g_{\text{in}}(v - Ax).$$

Then, there exists a classical solution $f \in \mathcal{C}^\infty\left([0,T]\times\mathbb{R}^6\right)$ with

$$f(t,x,v) = g(t, v - L(t)x), \qquad L(t) = A(I+tA)^{-1},$$

where $g(t,w)$ satisfies

$$\partial_t g = \langle L(t)w, \nabla_w g\rangle + Q_w(g).$$

Theorem 2: Hydrodynamic singularities

Let $\gamma \in [-3,\sqrt{22}]$, $g_{\text{in}}(\cdot)\geq 0$, $g_{\text{in}} \in C_c^\infty(\mathbb{R}^3)$ and initial data

$$f_{\text{in}}(x,v) = g_{\text{in}}(x+v), \qquad (A=-I).$$

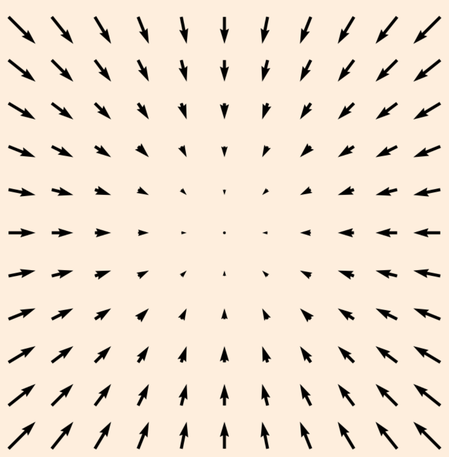
Then there is $f \in \mathcal{C}^\infty\left([0,1)\times\mathbb{R}^6\right) \cap L^\infty([0,1]\times\mathbb{R}^6)$ that develops hydrodynamic singularities at $t=1$:

$$\rho(t) = (1-t)^{-3}\int_{\mathbb{R}^3} g_{\text{in}}\, dw, \qquad e(t) = (1-t)^{-2}\frac{\int_{\mathbb{R}^3}|w|^2 g_{\text{in}}\, dw}{\int_{\mathbb{R}^3} g_{\text{in}}\, dw}.$$

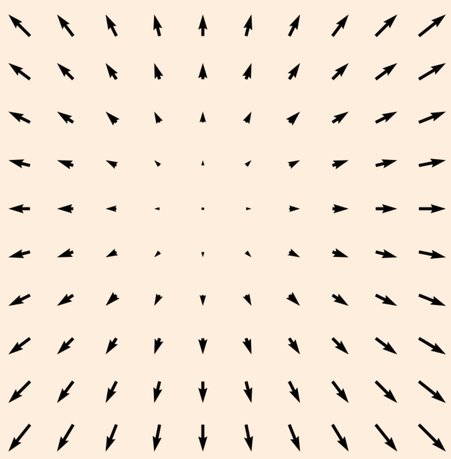
Moreover, f is written as

$$f(t,x,v) = g\left(s,(1-t)v+x\right), \qquad s := \int_0^t (1-\tau)^{-3-\gamma}\, d\tau,$$

where g solves the **homogeneous** Landau equation.



$u(t,x)$ for $A=-I$



$u(t,x)$ for $A=I$

Homoenergetic solutions

Introduce the macroscopic quantities $\rho(t,x)$, $u(t,x)$, $e(t,x)$

$$\rho = \int f\, dv, \qquad \rho u = \int v f\, dv, \qquad \rho e = \int |v-u|^2 f\, dv.$$

Main features of homoenergetic solutions are

$$\rho(t,x) = \rho(t), \qquad e(t,x) = e(t), \qquad u(t,x) = L(t)x.$$

Derived from the ansatz

$$f(t,x,v) = g(t,w), \qquad w = v - \xi(t,x),$$

which removes the variable x when $\xi(t,x) = L(t)x$, with

$$L'(t) = -L(t)^2 \implies L(t) = A(I+tA)^{-1}.$$

Other properties:

- Since $\rho(t,x)=\rho(t)$, we have $f(t) \in L^\infty_x L^1_v \setminus L^1_{x,v}$.
- If $A \neq 0$, there is no $C, m > 0$ such that $f_{\text{in}}(x,v) \leq C|v|^{-m}$. Therefore, [GHS,'26] does not apply.
- If $A \neq 0$, then f_{in} is not [HST,'20]-well-distributed.

Well-posedness for $-3 < \gamma < 0$

We work on the equation for $g(t,w)$.

Step 1. Estimate of mass $\rho(t)$, and moments $M_s(g)$.

$$\rho(t) = \frac{\rho(0)}{\det(I+tA)}, \qquad \sup_{t\in[0,T]} M_s(g) \leq C(T, M_s(g_{\text{in}}))$$

Step 2. Coercivity estimate. There is $\lambda(t) > 0$ with

$$\int |w-w_*|^\gamma \Phi(w-w_*) g(t,w_*)\, dw_* \geq \lambda(t)\langle w\rangle^\gamma I.$$

Step 3. Estimate on Fisher information based on [GS,'25]

$$\sup_{t\in[0,T]} \int |\nabla\sqrt{g}|^2\, dw \leq C\left(T, \int |\nabla\sqrt{g}_{\text{in}}|^2\, dw\right).$$

Step 4. Bound of L^p norms

$$\frac{d}{dt}\int g^p \leq -\lambda_T \int |\nabla g^{\frac{p}{2}}|^2 \langle w\rangle^\gamma + pC_{T,\gamma}\int (|w|^\gamma * g) g^p.$$

Step 5. $\left\||w|^\gamma * g\right\|_{L^\infty} \lesssim \|g\|_{L^{q_\gamma}}$ implies $g \in L^\infty$.

Self-similar solution

The solution from Theorem 2 is a particular case of the ansatz

$$f(t,x,v) = F\left(s(t), X, V\right), \qquad X := \frac{x}{(1-t)^{c_x}}, \quad V := \frac{v}{(1-t)^{c_v}}.$$

Picking $c_x = 0$, $c_v = -1$ we have that F solves

$$(1-t)s'(t)\partial_s F + V\cdot(\nabla_X - \nabla_V)F = (1-t)^{-2-\gamma}Q(F).$$

If we now assume $F(s,X,V) = G(s, X+V)$, we are left with

$$s'(t)\partial_s G = (1-t)^{-3-\gamma}Q(G),$$

which explains the time reparameterization in Theorem 2.